MILITARY SPECIFICATION

ACCELEROMETER, ELECTRICAL, LINEAR TRU-3/A

1. SCOPE

1.1 Scope. This specification covers a remote accelerometer, transmitter, -4 to +10g range, synchro output, designated TRU-3/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-C-5541	Chemical Films For Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited

STANDARDS

Military

DOD-STD-100	Engineering Drawing Practices
DOD-STD-1866	Soldering Process General (Non Electrical)(Metric)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Specifications And Standards, Order Of Precedence For The Selection Of
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of
MIL-STD-810C	Environmental Test Methods
MIL-STD-838	Lubrication Of Military Equipment
MIL-STD-889	Dissimilar Metals

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subject to first article inspection (see 4.4 and 6.4).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143.

3.3 Materials.

3.3.1 Fungus proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.3.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the accelerometer except where ferrous materials are essential.

3.3.4 Metals. Metals shall be of the corrosion-resistant type or suitably

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treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.6 Corrosive fumes. The materials as installed in the accelerometer shall not liberate deleterious fumes.

3.3.7 Protective treatment. When materials are used in the construction of the accelerometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.8 Recovered materials. Recovered materials shall be use to the maximum extent possible without jeopardizing the end use of the item.

3.4 Design and construction.

3.4.1 Design. The transmitter shall be designed to sense aircraft accelerations in maneuvers and rough air and shall conform to figure 1. The accelerometer shall meet the response characteristics specified in section 4.

3.4.1.1 Sensing. The mechanism shall sense vertical accelerations when the plans of the mounting lugs on the transducer is horizontal and when the accelerometer is mounted with the mounting lugs at the bottom. When in this position normal gravity shall be sensed at +1g. The transducer shall be designed to supply signals to a remote indicator so that the indicator shall exhibit accelerations in the range from -4 to +10g.

3.4.2 Construction. The construction of the accelerometer shall be reasonably simple to permit overhaul or repair without the use of special tools or fixtures. The accelerometer shall be so constructed as to withstand the normal strains, jars, vibrations, installation, and service, without failure.

3.4.2.1 Pivots, bearing and gears. Pivots, bearings, and gears shall neither bind nor shake and shall be as near frictionless as practicable.

3.5 Performance. The accelerometer shall be capable of meeting the tests specified herein when subjected to the following conditions.

a. Temperature. Temperature ranging from +150° to -54°C.

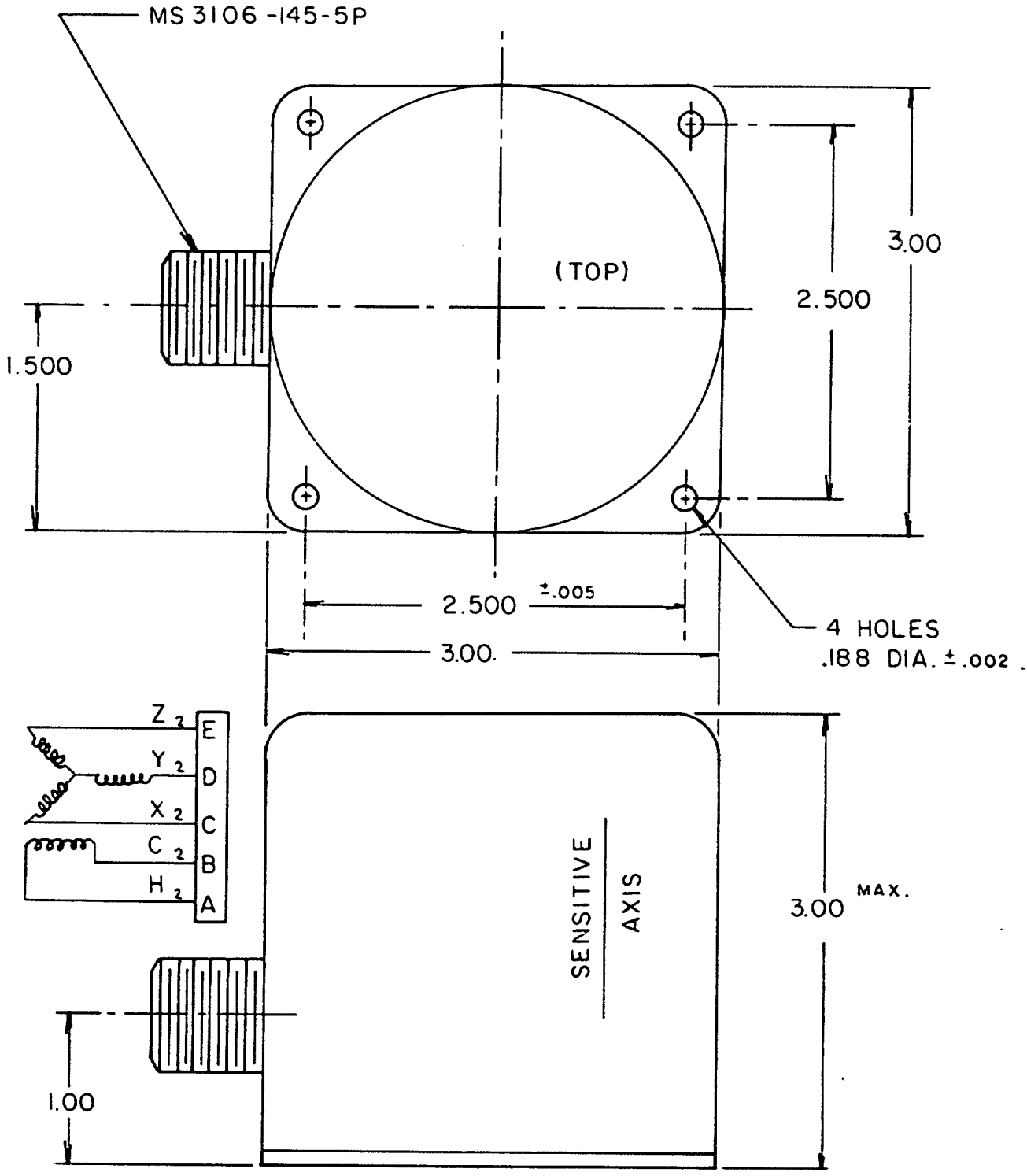
b. Humidity. Relative humidity up to 100 percent.

c. Altitude. Pressures ranging from 0 to 80,000 feet pressure-altitude.

d. Fungus. Fungus growth as encountered in tropical climates.

e. Thermal shock. Thermal shock with alternate immersion in water at temperature of +85° and +5°C for 8 cycles.

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED
TOLERANCES: DECIMALS ±.01

FIGURE 1. Case dimensions

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f. Sealing. The leak rate, based on atmosphere differential pressure and pure helium as the filling medium, shall not exceed 0.10 micron cubic foot per hour.

g. Acceleration and shock. Acceleration and shock forces as anticipated.

h. Vibration endurance. The steady state scale error shall not exceed 0.1g when subjected to the vibration endurance test.

i. Static scale error and friction. Static error and friction shall not exceed ±0.1g when tested as specified.

j. Dynamic scale error. Dynamic scale error shall not exceed ±0.1g.

k. Mounting lugs. The mounting lugs shall withstand a load of 175 pounds without fracture.

l. Response. Response characteristics shall be within the limits specified in 4.6.5 when the accelerometer is vibrated at frequencies from 3 to 20 Hz at maximum g forces for each test point.

m. Life. The accelerometer shall operate satisfactorily after being subjected to the frequencies and double amplitudes for the time periods specified in the life test of 4.6.16.

n. Salt fog. Exposure to salt sea atmosphere for a period of 48 hours.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and manufacturer's part numbers and changes thereto.

3.7 Case. The case shall be in accordance with figure 1 and the requirements specified herein. It shall provide a hermetically sealed inclosure for the mechanism of the instrument. The design of the case shall be such that the mechanism may be removed from the case, replaced and the case resealed without the use of special tools and fixtures unless they are approved by the contracting activity. The hermetic sealing shall be accomplished so that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the accelerometer may be subjected. The case shall meet the pressure requirements necessary when tested for leaks by means of a mass spectrometer type of helium leak detector as specified herein.

3.7.1 Filling medium. In the hermetically sealed case the filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligrams of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1/2 to 1 atmosphere as required. Where practicable the 100 percent helium filling medium shall be utilized.

3.8 Electrical connector. The electrical connector shall be MS3106-14S-5P for hermetically sealed mounting, shall be equipped with water-vaporproof cap and gasket, and shall be connected to the internal mechanism as follows:

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Pin	Internal Connection Synchro AY-503-53-A1
A	Single-phase winding No. 1 - C
B	Single-phase winding No. 2 - H
C	Three-phase winding No. 1 - Y
D	Three-phase winding No. 2 - Z
E	Three-phase winding No. 3 - X

3.9 Synchro transmitter output. The output shall be an Eclipse-Pioneer synchro type AY-503-8-A1 or electrical equivalent. The synchro shall rotate 10° per 1g. The synchro electrical zero shall be located at the 1g position and set at low null as specified in 4.3.8.

3.10 Acceleration-responsive element.

3.10.1 Adjustment. The mechanism shall be provided with means to adjust or correct the indications. This means of adjustment shall be reasonably simple to manipulate and of such a nature that adjustment can be accomplished with tools ordinarily possessed by an instrument repairman.

3.10.2 Stops for acceleration-responsive element. Mechanical stops shall be provide for arresting the motion of the acceleration-responsive element when the accelerometer is subjected to accelerations beyond the limits specified. One stop shall engage the element at an acceleration of -4.3g $\pm$0.2g and the other at an acceleration of +10.3g $\pm$0.2g.

3.11 Power. The accelerometer shall be designed to operated on 26V single phase 400 Hz as excitation of the synchro only.

3.12 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454.

3.13 Lubrication. The accelerometer shall be lubricated in accordance with MIL-STD-838.

3.14 Weight. The weight of each accelerometer shall not exceed 2 pounds.

3.15 Finishes and protective coatings.

3.15.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.15.1.1 Small holes and case inserts. Small holes, and case inserts need not be anodized.

3.15.1.2 Aluminum alloys. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-G-5541.

3.15.1.3 Paint base. Where the the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.15.1.4 Castings. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

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3.15.1.5 Abrasion resistance. When abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.15.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488,Type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating in lieu of MIL-C-83488, may be use if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternate processes can be used.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship. The accelerometer, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner in accordance with MIL-STD-454, requirement 9. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, and bolts.

3.17.1 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads.

3.17.2 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.17.3 Gears. Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots or other irregularities. Where required, for accurate adjustment, gear assemblies shall be free from backlash.

3.17.4 Cleaning. The accelerometer shall be thoroughly cleaned of loose, spattered or excess solder, metal chips or other foreign material after assembly. Burrs and sharp edges as well as resin flash which might crumble shall be removed.

3.17.5 Dimensions. Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the accelerometer they shall be held to or limited accordingly.

3.18 Envelope. An envelope, containing the mounting screws, shall be furnished with each accelerometer. The following information shall be printed on the face of the envelope:

"IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS"

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purhcase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of tests. The inspection requirements specified herein are classified as follows:

a. First article inspection test (see 4.4).

b. Acceptance tests (see 4.5).

4.3 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test conditions specified in 4.3.1 of this specification. No adjustments or repairs of any type shall be performed on the samples after start of first article.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Tapping. Unless otherwise specified, the accelerometer shall be tested in the normal operating position and shall be gently tapped or vibrated before a test reading is taken.

4.3.3 Standard test indicator. Facilities for determining the rotor position of the follow-up synchro to within $\pm 0.1^{\circ}$ shall be provided and the servo system shall not introduce more than 0.1° error in the indication. The single phase winding of the transmitter synchro shall be excited with 26V, 400 Hz, and the error signal for the amplifier shall be obtained from the single phase winding of the follow-up synchro. The indicator shall be set to indicate 10° when connected to the transmitter at the electrical zero. For tests where it is desired to record a transient condition an oscillograph may be substituted for the indicator.

4.3.4 Identification of synchro leads. The following procedure shall be used for identifying the synchro leads:

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a. Except for use of the ZY 503-53-A1 synchro, the rotor leads shall be arbitrarily labeled H and C and one of the stator leads shall be arbitrarily labeled Z. Lead Z shall be connected to lead C. The two unlabeled stator leads shall be shorted together and a vacuum tube voltmeter (VTVM) shall be connected between the two unlabeled stator leads and the rotor lead H. A voltage of 26V, 400 Hz single phase excitation shall be applied between leads H and C. The voltage measured on the VTVM should be less than the rotor excitation voltage (approximately 16V).

b. If the voltage measured in a. above is greater than, or approximately equal to the rotor excitation voltage, repeat the procedure except that the rotor shaft shall be rotated 180° from its previous position until the desired null voltage is measured. Then the synchro rotor is at the correct electrical zero position.

c. With the synchro set at the position as determined in b. above, measure the voltage between either of the unlabeled stator leads and lead H. If the rotation of the synchro rotor shaft in a clockwise direction (when viewed from the shaft end) causes the measured voltage to increase before decreasing, the VTVM is connected between leads H and X. If the measured voltage is caused to decrease before increasing, the VTVM is connected between leads H and Y.

d. Pin notations should be as assigned in 3.8.

4.3.5 Determination of approximate electrical zero position. The following procedure shall be used:

a. Connect pin A to pin D

b. Connect pin C to pine E and on side of a voltmeter

c. Connect pin B to the other side of the voltmeter

d. Apply 26V, 400 Hz to pins A and B.

The synchro shaft or housing is then rotated to either of two positions which are 180° apart. The correct approximate position is that one which procedures the smaller voltage (approximately 16V).

4.3.5.1 Determination of exact electrical zero position. The following procedure shall be used:

a. Connect pin C to one side of a VTVM and pin E to the other side of the VTVM

b. Apply 26V, 400 Hz to pines A and B

c. Rotate the synchro housing or shaft through the smaller angle that will produce a low null reading on the VTVM. This is exact electrical zero position.

4.4 First article inspection. The first article inspection shall consist of all tests described under 4.6.

4.4.1 Inspection samples. The first article inspection samples shall consist of

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3 accelerometers representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the contracting activity.

4.5 Acceptance inspections. Acceptance inspections shall consist of:

a. Individual inspections

b. Sampling plans and inspections

4.5.1 Individual inspections. Each accelerometer shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Static scale error and friction

c. Dynamic scale error

d. Sealing

e. Response characteristics

4.5.2 Sampling plans and inspections.

4.5.2.1 Sampling plan A inspection. One accelerometer selected at random from each 100 or less product on the contract or order shall be subjected to the following tests as described under 4.6:

a. Individual inspections

b. Low temperature

c. High temperature

4.5.2.2 Sampling plan B inspection. Unless otherwise specified, 3 accelerometers selected at random from the first 15 items of the contract or order shall be subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Vibration endurance

c. Shock

d. Mounting lugs

e. Temperature - altitude

f. Humidity

g. Fungus

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h. Thermal shock

i. High temperature exposure

j. Life

k. Dielectric

l. Salt fog

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. Each accelerometer shall be examined externally to determine conformance with this specification with respect to material, workmanship, marking, and envelope.

4.6.2 Static scale error and friction. The position error shall be observed with the accelerometer held in each of the following positions:

a. Plane of mounting lugs horizontal and below the case.

b. Plane of mounting lugs vertical.

c. Plane mounting lugs horizontal and above the case.

In each position the test indicator reading obtained shall be due entirely to gravity and not to motion in change of position of instrument. The static scale error shall not exceed $\pm$0.1g in any of the three positions tested. The accelerometer shall also be tested for friction at the three positions mentioned above. The amount of tapping or vibration applied shall be sufficient to remove all friction from the accelerometer. After the desired test position is reached two readings shall be taken, the first before vibration and the second after vibration. The difference of two such readings shall be recorded as friction and shall not exceed 0.1g in any of the three positions tested.

4.6.3 Dynamic scale error at room temperature. The accelerometer shall be mounted on a suitable centrifuge and the scale errors determined for all accelerations specified in table I. At room temperature the scale error shall not exceed 0.1g at any test point.

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4.6.4 Sealing. The accelerometer case shall be placed in a chamber and tested for leaks by means of a mass-spectrometer type of helium leak detector. The chamber shall be evacuated so that the differential pressure existing when the test is conducted is approximately the same as that to which the accelerometer case is filled. The leak rate shall be based on 1 atmosphere differential pressure and pure helium as the filling medium. Under these conditions the leak rate shall not exceed 0.10 micron cubic foot per hour. Equivalent helium leak rates for the same hole size at other differential pressures and helium-nitrogen mixtures, or both, shall be calculated by the manufacturer.

TABLE I.

Dynamic Scale Error		
Input Acceleration In "g"s	Scale Error Tolerance $-54^{o}C$	Scale Error Tolerance $+150^{o}C$
-4	0.2	0.2
-3	0.2	0.2
-2	0.1	0.1
-1	0.1	0.1
0	0.1	0.1
1	0.1	0.1
2	0.1	0.1
3	0.2	0.2
4	0.2	0.2
5	0.2	0.2
6	0.3	0.2
7	0.3	0.2
8	0.3	0.2
9	0.4	0.3
10	0.4	0.3

4.6.5 Response characteristics. The transmitter shall be subjected to vibration along the vertical axis. The test apparatus shall be as specified in 4.6.5.1. The transmitter shall be tested at frequencies of 3, 6, 9, 12, 15, 18, and 20 Hz. The amplitude of vibration at each test point shall be such that the maximum g-forces practicable for that test point will be obtained. The ration of indicated acceleration to applied acceleration shall be in accordance with Table II.

4.6.5.1 Test apparatus. The equipment shall impose linear sinusoidal motion to the accelerometer and shall be capable of producing the frequencies and double amplitude required for each test point. The acceleration factor of any harmonic measured on the test table shall not exceed 10 percent of the fundamental acceleration vector.

4.6.6 Low temperature. The accelerometer shall be subjected to the low temperature test in accordance with Method 502.1, procedure I of MIL-STD-810C for a period of 4 hours in lieu of 24 hours. Upon completion of the test period and while at the test temperature the accelerometer shall be subjected to meet the requirements of 4.6.2 and 4.6.3.

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TABLE II. Response Characteristics

Frequency in Hzs	Ratio Indicated g/Applied g
3	0.94 to 1.06
6	0.56 to 0.82
9	0 to 0.45
12	0 to 0.26
15	0 to 0.18
18	0 to 0.13
20	0 to 0.10

4.6.7 High temperature. The accelerometer shall be subjected to the high temperature test in accordance with Method 501.1, procedure I or MIL-STD-810C for a period of 4 hours in lieu of 48 hours and 121°C in lieu of 71°C. Upon completion of the test period and while at the test temperture the accelerometer shall be subjected to and meet the individual tests.

4.6.8 Vibration endurance. The accelerometer shall be subjected to the vibration test in accordance with Method 514.2, procedure I, part I, category b.1 equipment (curve b) of MIL-STD-810C. The range of curve b shall be from 5 to 50 Hz. Sinusoidal cycling only will be required, except the period shall be 50 hours in lieu of 3 hours. Upon completion of the test period the accelerometer shall be subjected to the test specified in 4.6.3. The steady state scale error shall not exceed 0.1g.

4.6.9 Shock. The accelerometer shall be installed in the normal operation position in a free-fall apparatus as described in 4.6.9.1. The accelerometer shall be allowed to fall from a height of 8 inches, then examined for damage and subjected to the test specified in 4.6.3. There shall be no damage to the accelerometer and the steady state scale error scale shall not exceed ±0.1g.

4.6.9.1 Apparatus. A rigidly constructed device provided with a panel mounting for installing the accelerometer in the specified position, and vertical guides to allow free-fall from the specified height shall be used. A bumper on the free-fall element shall arrest its fall by striking the center of a rounded projection on a metal block. The bumper shall be spring steel .25 inch by 1.5 inches in cross section with an unsupported length of 5 inches between points of attachment to the free-fall elements. The bumper shall contact the metal block at the midpoint of a broad side. The center of gravity of the instrument under test and the contact point of the bumper shall be aligned vertically within .125 inch. The weight of the free-fall element shall be approximately 12 pounds. The metal block shall weigh not less than 50 pounds.

4.6.10 Mounting lugs. The accelerometer case with mechanism removed shall be mounted on the movable hand of a suitable testing machine with the diameter of the case in a horizontal plane so the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied along the axis of the mounting hole for a period of 1 minute. There shall be no fracture.

4.6.11 Temperature-altitude. The accelerometer shall be subjected to the test

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in accordance with procedure I, equipment category 7 of MIL-STD-810C. The time duration for step 1b and step 2 of table 504-1-II shall be 48 hours and 24 hours respectively. Perform, only steps 1b, 2, and 3. There shall be no damage to the accelerometer as a result of this test.

4.6.12 Humidity. The accelerometer shall be subjected to the humidity test in accordance with Method 507.1, procedure I of MIL-STD-810C.

4.6.13 Fungus. The accelerometer shall be subjected to the fungus resistance test in accordance with Method 508.1, procedure I of MIL-STD-810C.

4.6.14 Thermal shock cycling. The accelerometer shall be alternately subjected to a total of 8 cycles of immersion in tap water maintained at 85°C ±2°C and 5° ±2°C. The length of time for each immersion shall be 30 minutes; not more than 5 seconds shall elapse between immersions. No evidence of moisture penetration or damage penetration or damage to the inclosure shall result from this test. Following this test, the accelerometer shall be subjected to and meet the requirements for the test specified in 4.6.4.

4.6.15 High temperature exposure. The accelerometer shall be subjected to the high temperature test in accordance with Method 501.1, procedure I of MIL-STD-810C except that the temperature shall be 150°C. No damage to the accelerometer shall result form this test.

4.6.16 Life. The accelerometer shall be mounted on a device as described in 4.6.16.1. It shall be subjected to the frequencies and double amplitudes for the times specified and in the order specified in table III.

4.6.16.1 Accelerometer. The equipment shall impose linear sinusoidal motion to the accelerometer and shall be capable of producing the frequencies and double amplitudes (±1/32 inch) required by table II. The acceleration vector of any harmonic measured on the test table shall not exceed 10 percent of the fundamental acceleration vector.

4.6.17 Dielectric. The accelerometer shall be subjected to a dielectric test with 100V dc applied between each terminal and the case of the accelerometer for a period of 1 minute. The resistance shall not be less than 20 megohms.

4.6.18 Salt fog. The accelerometer shall be subjected to the salt fog test in accordance with Method 509.1, procedure I of MIL-STD-810C. Upon completion of the 50-hour test period, there shall be no evidence of corrosion as a result of this test.

4.7 Inspection of packaging. The inspection of the preservation-packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements or MIL-STD-129.

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TABLE III.

Life Test			
Impressed Frequency CPS	Double Amplitude Inches	Applied Acceleration g	Time Hours-Minutes
1	10	1.02	1 - 40
2	10	4.08	50
3	8	7.35	33
3	6	5.51	33
4	4	6.53	25
5	2	5.10	20
6	2	7.35	17
6	1	3.68	17
7	1	5.00	14
8	1	6.53	12

NOTE: Time is figured so that at each frequency, approximately 6000 cycles are applied or a total of approximately 60,000 cycles.

5 PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Transmitter shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method 1C1 of MIL-P-116. Overbox complete pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

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5.2.1 Level A. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2 Level B. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type accelerometer covered by this specification is intended for use on aircraft to sense vertical acceleration in maneuvers and rough air.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Whether sampling plan B is to be omitted

c. Reinspection of date markings

d. Level of packaging and packing desired.

6.3 Definitions.

6.3.1 Tapping. The phrase TAPPED LIGHTLY is defined as meaning the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.3.2 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.3 g Unit. The g unit is defines as an acceleration of 32.2 feet per second.

6.3.4 Amplitude. Whenever the word AMPLITUDE is specified, it shall mean that extent of motion as measured from one extreme to the opposite extreme.

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6.4 First article. When first article inspection is required sample units shall be inspected as specified (see 3.1 and 4.4).

6.5 Subject term (key word) listing.

Accelerometer
Hermetically Sealed
Linear
Range-4 to 10g
Remote
Transmitter

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility no any obligation whatsoever; and the fact that the government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
AIR FORCE - 99

Preparing Activity:
AIR FORCE - 71

Project Number:
6610 - F138

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-27330B	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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